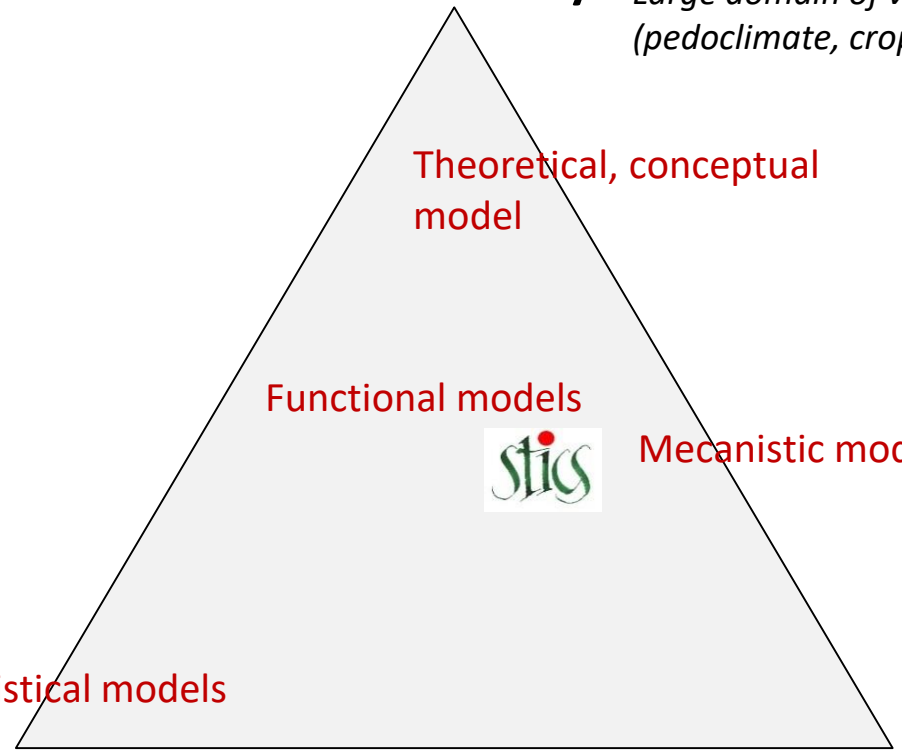
Modelling (and performance) is always a matter of compromise…

Genericity

Large domain of validity (pedoclimate, crop…)



+ 4th dimension feasibility ?

Precision

Accuracy, fidelity to observations: data-driven, focus on estimation error in a given context

Realism

Mechanisms, causal links, coverage of all processes

Levin's triangle for the types of model Levins, 1966

Where do the uncertainties come from?

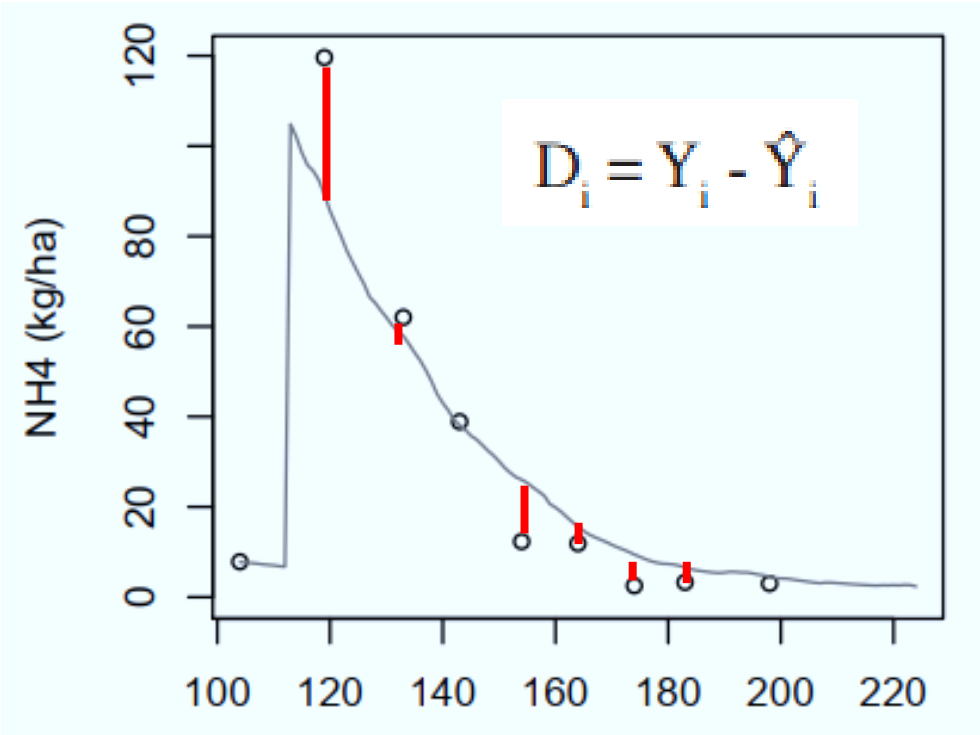
- **Different sources**
  - **Model itself :** choice of equations, pools and fluxes represented, scope of the model…
  - **Input Variables** : measurement errors, lack of information characterization, …
  - **Parameters of the equations** (we often tend to overemphasize this component of error)
  - **User**!

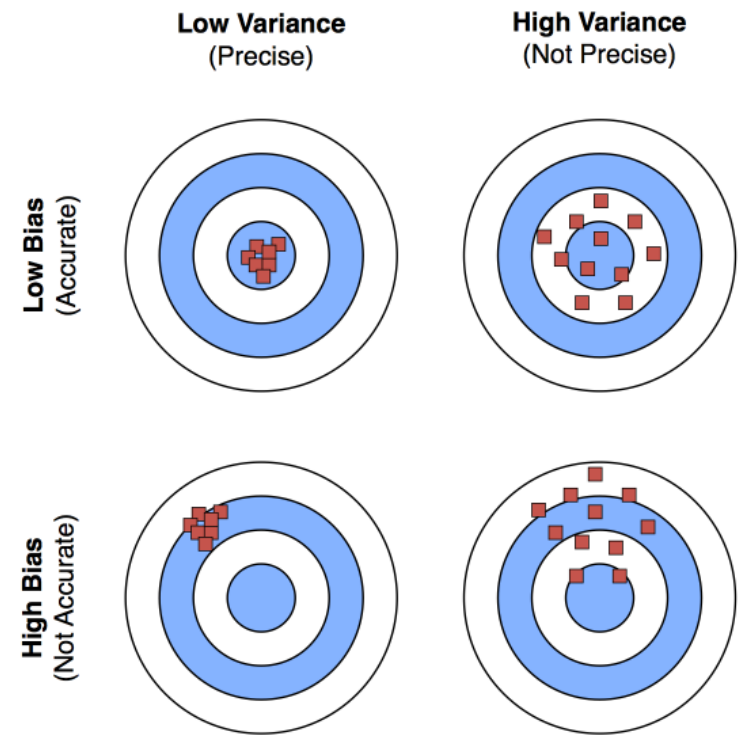
- **Confidence is strongly related to the domain of relevance/applicability of the model**
  - which depends on the range of situations over which the model has been built and tested (ex fertilization)
  - which depends on if some important phenomenon are not simulated by the model (ex : pests and diseases)

important to understand that a very good evaluation in one context does not imply accuracy (may be the contrary)

How models are evaluated?

Generally, **quantitative** assessment of **deviations from observations**





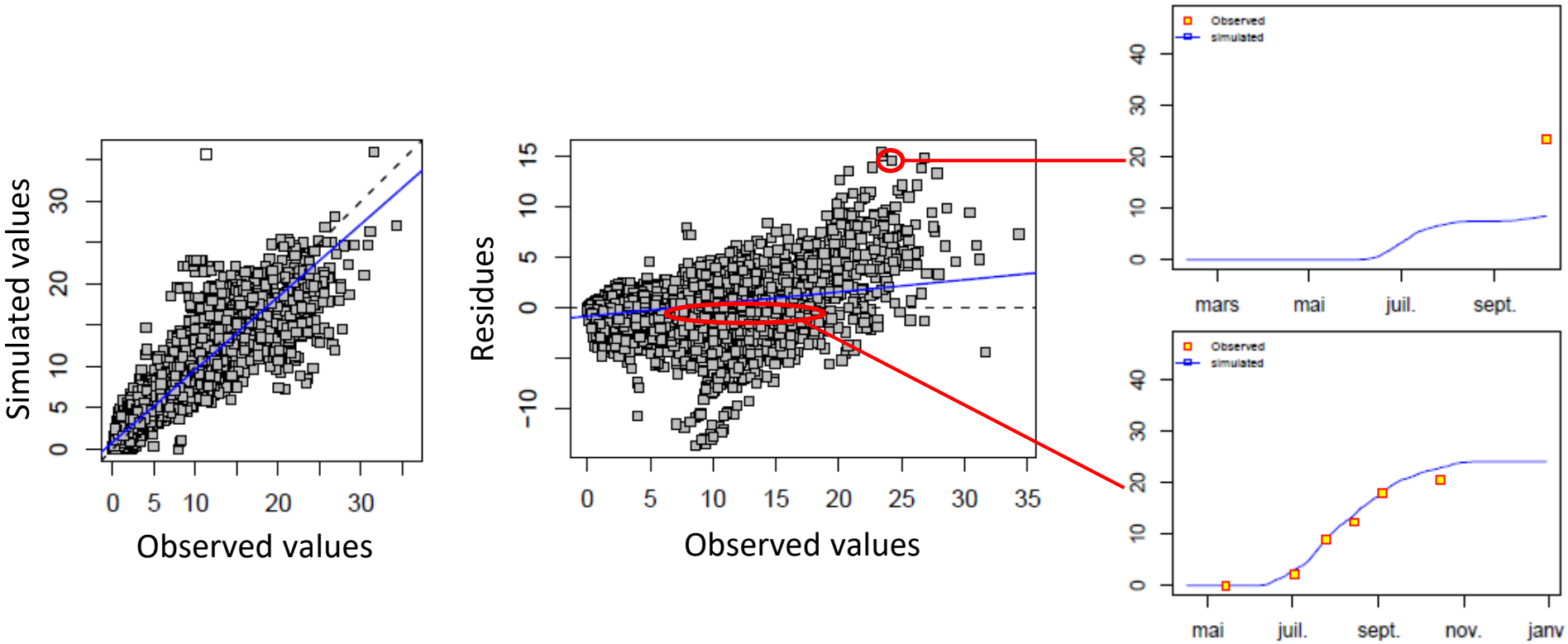
Sebastian Raschka

- /!\ possible error also on the measurement!
- difficulties: ex. time lags
- The deviation from observation is a source of knowledge!

Methods: Graphical plots comparing observations and simulations

- assess « qualitatively » the agreement between model results and observed data points

Much more informative than any statistical criteria!



Methods: Goodness-of-fit criteria

Computation of **numerical metrics** comparing model simulations to corresponding observations

 The most popular way of performing model evaluation, because they are quantitative and **seem** objective

- To me, the main interest is they can provide a **synthetic view**, facilitate **comparisons**, can be **used as numerical criteria** for automatic optimization.
- **I do not agree they are more objective**, and although « quantitative », they are **often much poorer than plots** (ex regression with outliers)
- /!\ performance in **fitting vs prediction**!

Goodness-of-fit criteria: examples

$$D_i = Y_i - \hat{Y}_i$$

- Difference (D_i) between obs. (Y_i) et sim. ($\hat{Y}_i$)

$$\text{Biais} = \frac{1}{N}\sum_{i=1}^{N} D_i$$

- Bias = mean of errors

Does not tell anything about dispersion

$$r = \frac{\sigma_{Y\hat{Y}}}{\sigma_Y \, \sigma_{\hat{Y}}}$$

- Pearson's coefficient of correlation (or r^2)

Together with slope and intercept to account for bias!

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N} (D_i)^2} = \sqrt{\frac{1}{N}\sum_{i=1}^{N} (Y_i - \hat{Y}_i)^2}$$

- RMSE (and RRMSE = $\frac{RMSE}{\frac{1}{N}\sum_{i=1}^{N} \widehat{Y_i}}$) Same unit

+ decomposition in bias and dispersion components

$$EF = 1 - \frac{\sum_{i=1}^{N} (Y_i - \hat{Y}_i)^2}{\sum_{i=1}^{N} (Y_i - \overline{Y})^2}$$

- Model efficiency

How far from the 1:1 line

- Include bias
- 1 = perfect, 0 = no better than average of obs

Goodness-of-fit criteria: references

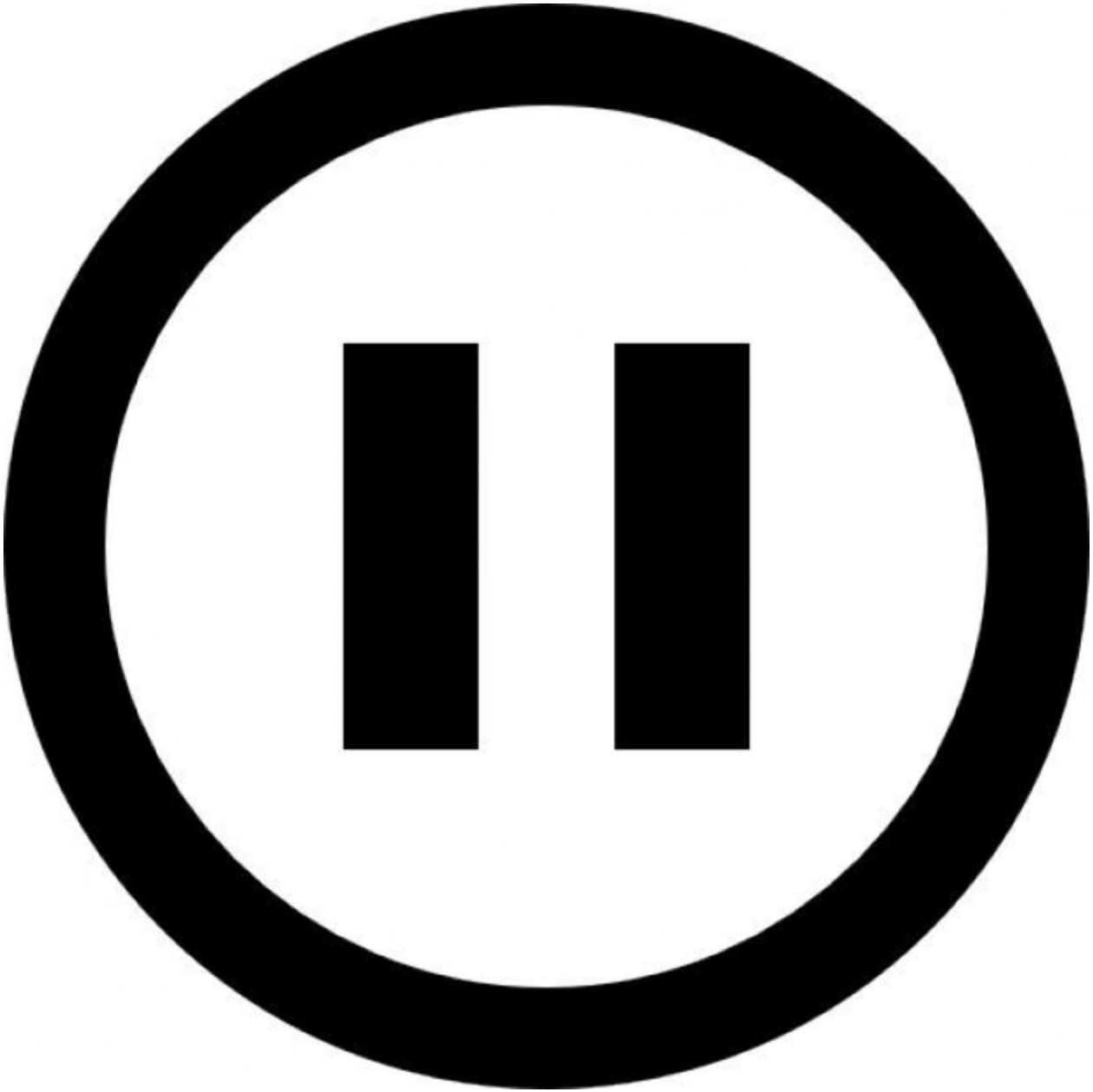
For a more exhaustive list of goodness-of-fit criteria with their description, see for example:

Wallach, D. (2006) Evaluating crop models. In: Working with Dynamic Crop Models: evaluation, analysis, parameterization and applications.

Bellocchi, G., et al. (2010). Validation of biophysical models: issues and methodologies. A review. *Agronomy for Sustainable Development*

Bennett, N. D., et al. (2013). Characterising performance of environmental models. *Environmental Modelling & Software*

Don’t spend too much time playing with criteria, your time is better used looking at plots!



Beyond goodness of fit criteria: behavioural analysis

- **Sensitivity analysis** (to changes in parameters, input values…)

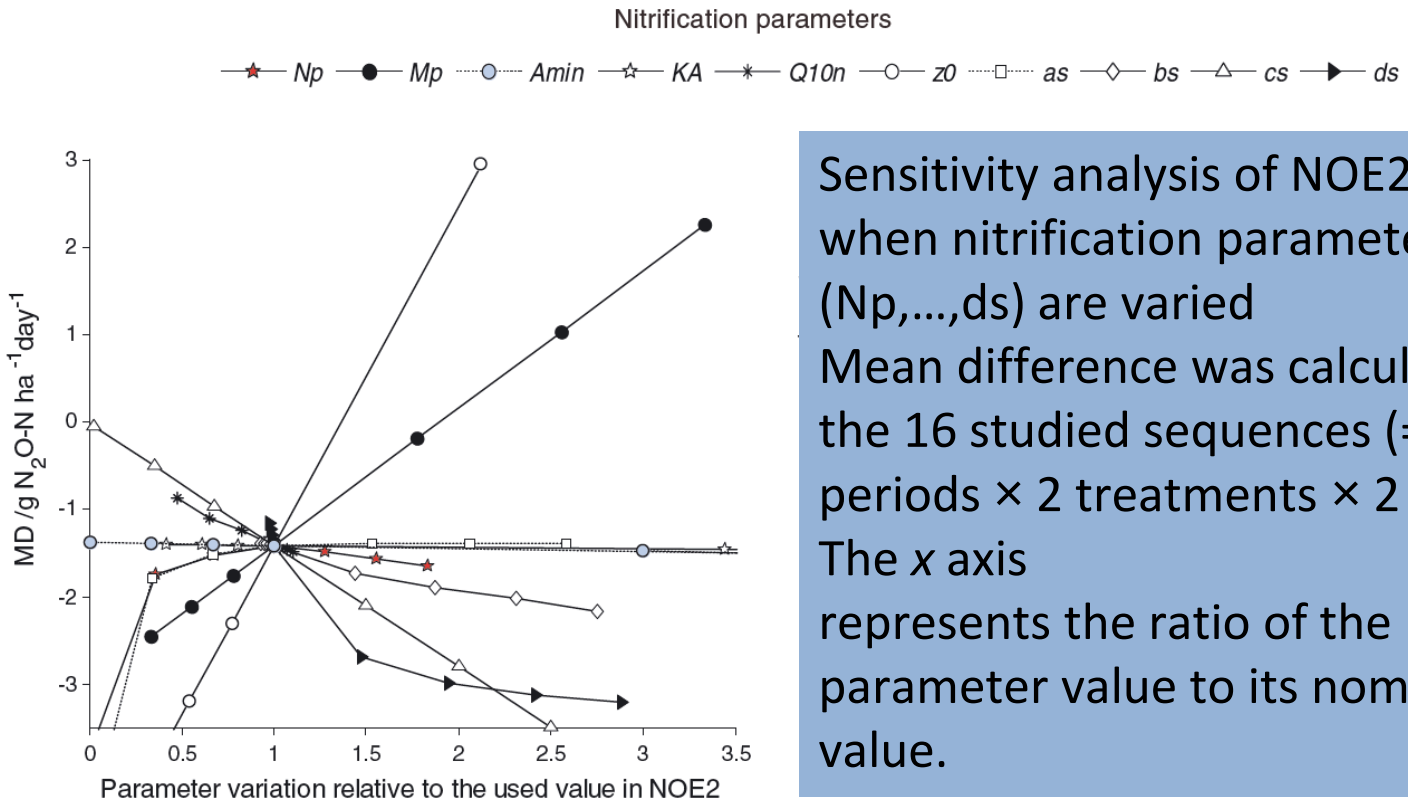
  not really a tool for evaluation of model performance, but of how the model behaves -> useful for better use of the model, calibration, to focus on key parameters

- **Analysis of the response** as a function of climatic variables, soil variables, management variables

  does the model behave as expected? Does the error change?

Local sensitivity analysis

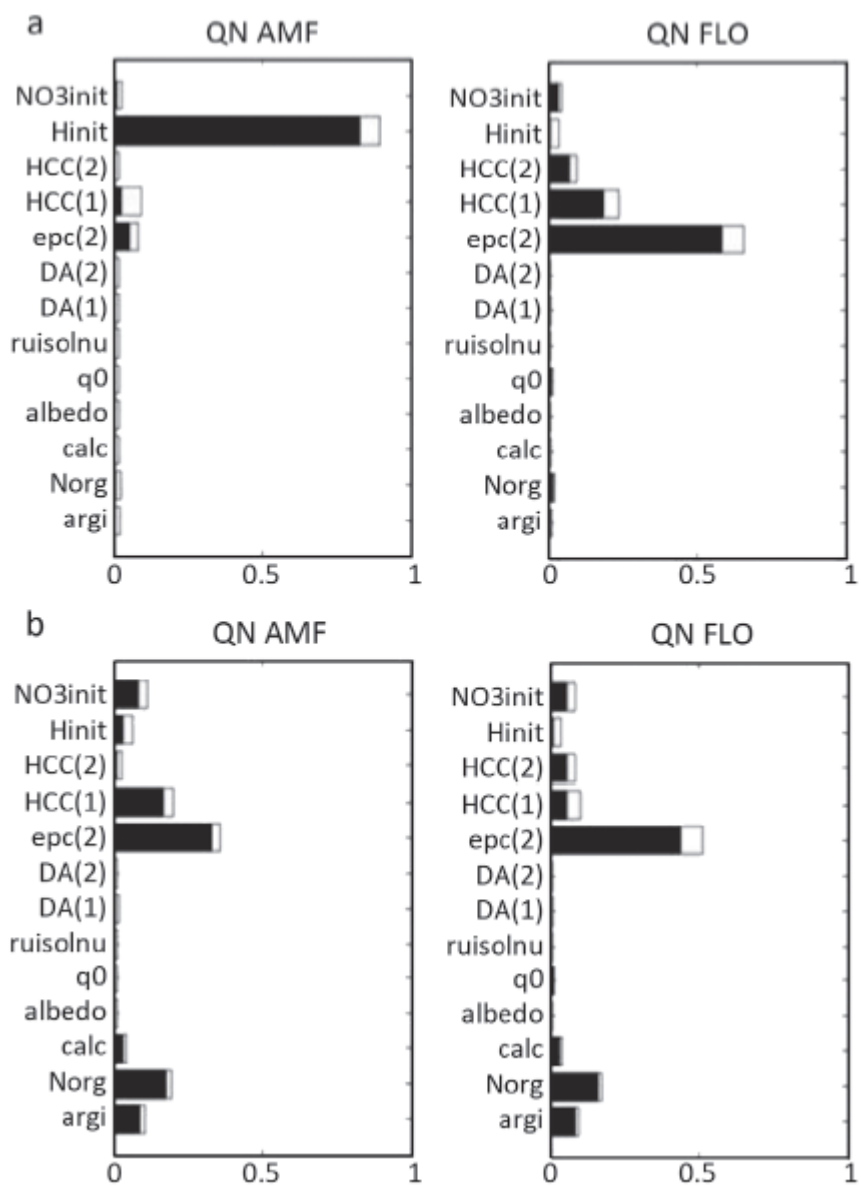
Example of the NOE model (STICS module) (Bessous et al., 2010)



Sensitivity analysis of NOE2 model when nitrification parameters (Np,…,ds) are varied
Mean difference was calculated for the 16 studied sequences (= 4 periods × 2 treatments × 2 years).
The *x* axis
represents the ratio of the parameter value to its nominal value.

- Be careful, a same relative variation may be a small or strong change…
- Can depend on conditions! Ex no response to denitrification potential if WFPS <0.62…

Global sensitivity analysis



Global sensitivity analysis study of STICS to soil properties (Buis et al., 2011 ; Varella et al, 2010)

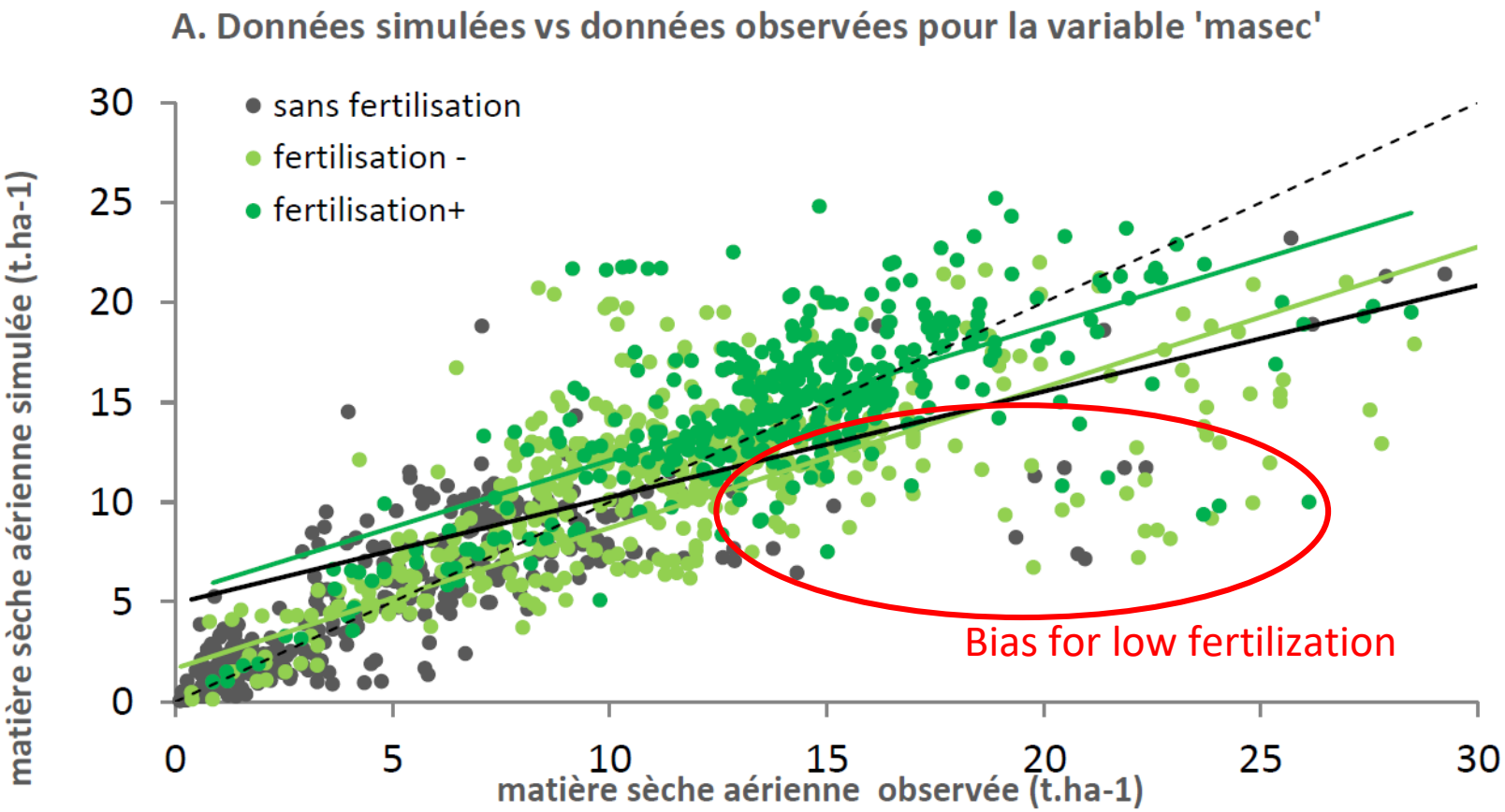
-> allow to study interactions
-> explore all parameter space, but heavy experimental design!

Fig. 14–6. Sensitivity indices of soil characteristics for the output variable aboveground plant N (QN) at beginning of stem elongation (AMF) and at flowering (FLO) for (a) dry climate and (b) wet climate. The first-order effects are represented in dark and interaction effects in light.

From (Buis et al, 2011)

Response as a function of different variables (climatic, soil, agronomic…)

Yield prediction accuracy as a function of Ninput (Coucheney et al; 2012)



Guidelines for model evaluation

- **The evaluation dataset should be representative of the conditions on which the model will be applied** (climate, soil, crops, management… although we expect some genericity)

  ⇒ *however this is seldom possible and a challenge for innovative systems!*

- For an objective/useful evaluation, **the evaluation dataset should not have been used to calibrate the model** (at least the part of the model we want to evaluate) : fitting is not predicting, and overfitting compromises genericity!

- **Each individual goodness-of-fit criteria only assess a single component of model behaviour** -> possibly good value for one and bad for another.

  → *Use of multiple complementary statistics is advised* ***(do not forget plots !!)***

/!\ Models are never « validated ». We can only continue to challenge the model and try to invalidate it.

How to deal with bad performances?

- Are the *__key__* processes for the situation considered included in the model?
  never try to solve bad performance by adding more processes if you do not have very good reasons to do that!

- check **dataset quality**: uncertainty on model inputs and observations (ex. Nmin very variable)

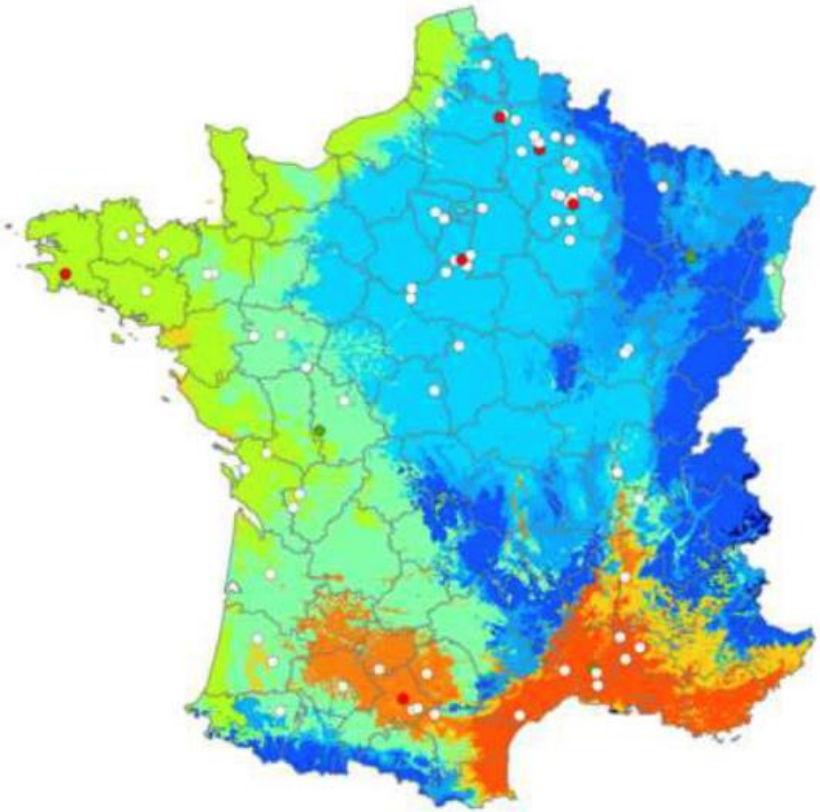
- Check the calibration process: are the data used **representative** of the studied case? Has performance in **prediction** evaluated?

- Share your problem on the Stics forum
  (provide good information about the system under study and the problem!)

An example of a multi-method evaluation

Coucheney et al., 2015, EMS

Multicriteria and multisite evaluation of Stics V8.2.2



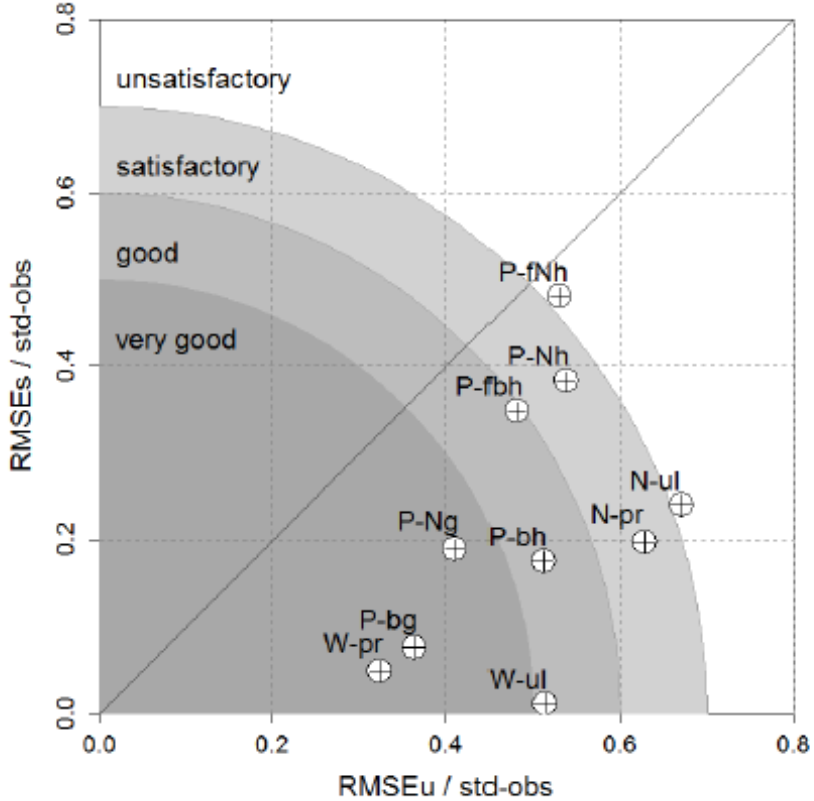
French datasets
15 crops + bare soil,
76 sites
440 sites.year
1978-2009
 1809 USMs

Evaluation for 10 observed variables biomass, nitrogen, soil, yield…

Standard set of parameters!

An example of a multi-method evaluation

Quantification of model accuracy



RMSE (s or u) = root mean square error (systematic or unsystematic) of simulations.
Std = Standard deviation of observations

Soil water and nitrate variables

- W-ul: water content in g g−1(upper layer)
- W-pr: water amount in mm (profile)
- N-ul: nitrate amount in kg ha−1 (upper layer)
- N-pr: nitrate amount in kg ha−1 (profile)

Plant variables at harvest

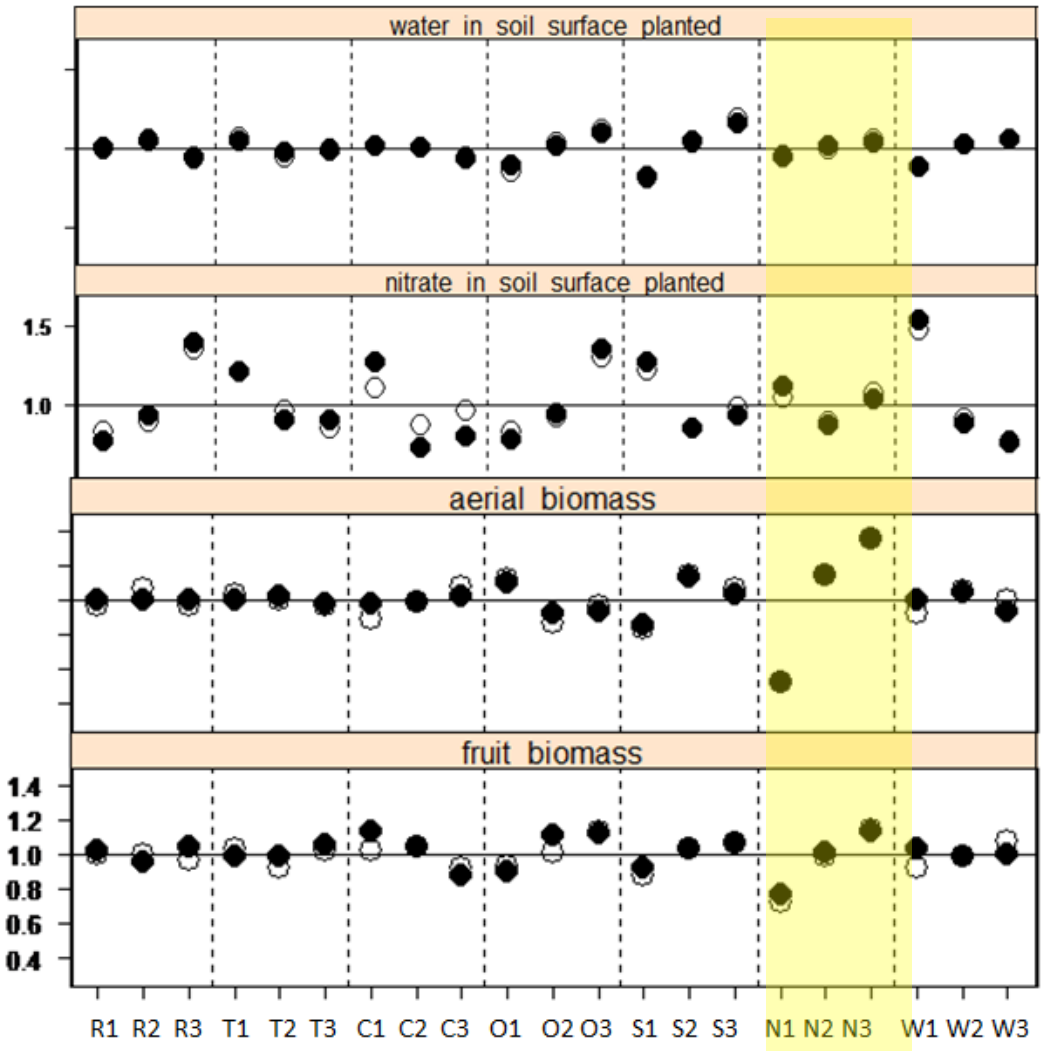
- P-bh: aerial biomass in t ha−1
- P-Nh: N in the plant in kg ha−1
- P-fbh: fruit biomass in t ha−1
- P-fNh: N concentration in fruits in % odb

Plant variables along the growing season

- P-bg: aerial biomass in t ha−1
- P-Ng: N in the plant in kg ha−1

An example of a multi-method evaluation

Behavioural analysis



Relative variations of simulated (open symbols) and observed (filled symbols) variables between classes of agropedoclimatic indicators (1 = low, 2 = moderate, 3 = high)

(i) climatic data: mean daily radiation (**R**) and frequency of minimal temperatures below 0°C (**T**);

(ii) soil data: clay content (**C**), organic nitrogen content (**O**) and water stock at field capacity (**S**);

(iii) available resource data: N fertilization rate (**N**) and mean daily water availability (**W**).

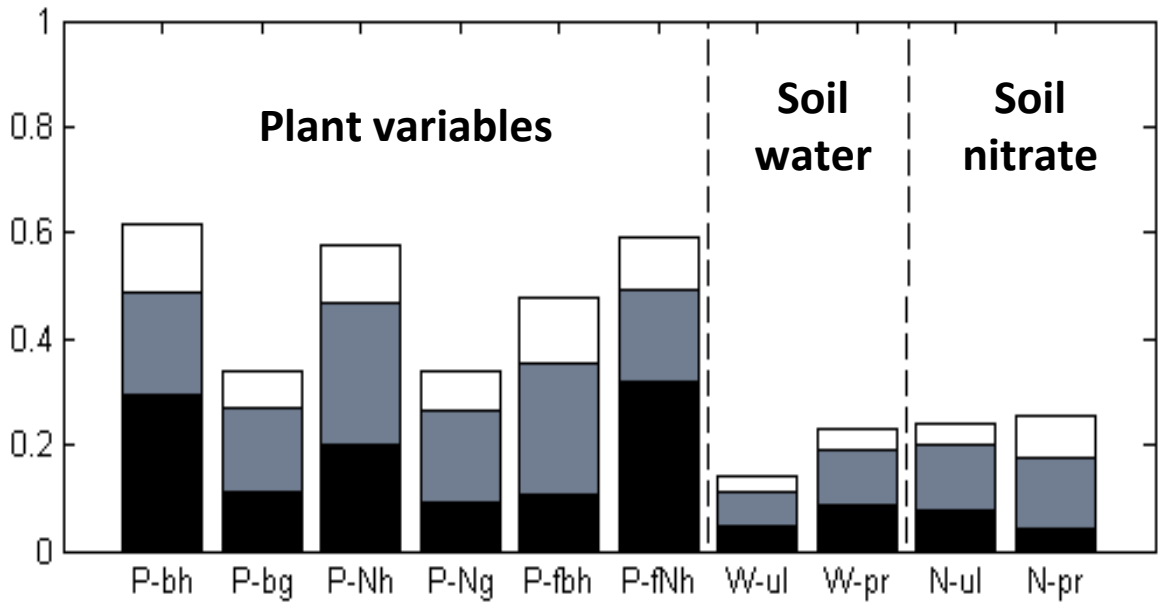
Does the model respond in the expected way to climate, management?

An example of a multi-method evaluation

Model robustness analysis

Percentage of variance of model residuals explained by:

- crop type (black bars),
- agropedoclimatic indicators (white bars, sum of their main effects and second order interactions)
- second order interactions (grey bars).

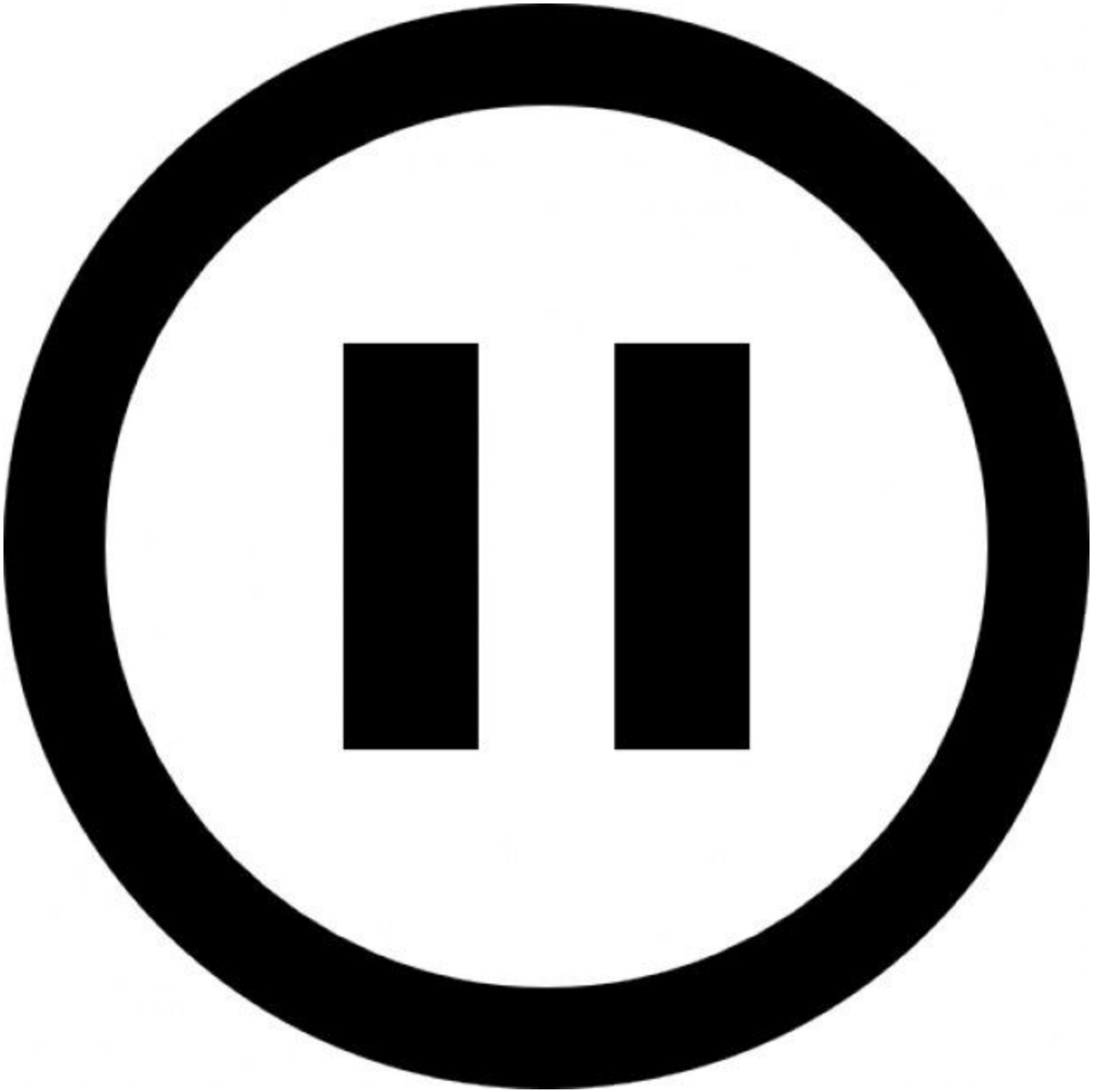


Not robust if explanation of error variance is dominated by a variable such as the crop type

An example of a multi-method evaluation

Main conclusions about Stics version V8.2.2 multi-method evaluation:

- Good overall accuracy, with little bias.
- Relative RMSE was larger for soil nitrate than for plant biomass and nitrogen and smaller for soil water.
- Trends induced by contrasted environmental conditions and management practices globally well reproduced.
- Similar performances on calibration and evaluation datasets.
- STICS more robust with respect to the main agropedoclimatic factors driving its processes than to the crop type.



INRAG

Calibration

Model parameters can be estimated from measurement, from litterature, physical/biological value, lookup tables/pedotransfer functions…

But in some cases none of these ways allows to get the parameter value(s), or that value does not give satisfactory results…

=> **Calibration**: iterative procedure to obtain parameter values that optimize simulated results so that they better fit the corresponding measured data

Calibration begins with parameter selection

- **Identify which parameters can be fixed** using scientific literature, extrapolation from other STICS plant files (i.e similar crop), measurements…
- Among the remaining, select the **most influential** ones (sensitivity analysis and expert knowledge)

Sets of parameters to calibrate depending on calibration type (crop, cultivar), data availability, … are proposed in chapt 16 of the book "STICS Soil-Crop Model, Conceptual Framework, equations and uses"

Example : Varietal calibration schemes, in the case of a determinate plant, without *.lai files available but with observed stages dates

Step	Forcing	Process	Chosen Settings	Targeted Variable
1	Pheno stages	Roots growth	*croirac*	zrac or défaut resmes
2		Biomass growth	sea	masec
3		Elaboration of yield	*nbgrmax, pgrainmaxi*	mafruit
4	Emergence date	Development	*stlevamf, jvc(annuelle), sensiphot*	IAMF
			stamflax, stlevdrp	ILAX, IDRP
6		Foliage growth	*adens, durvieF*	LAI

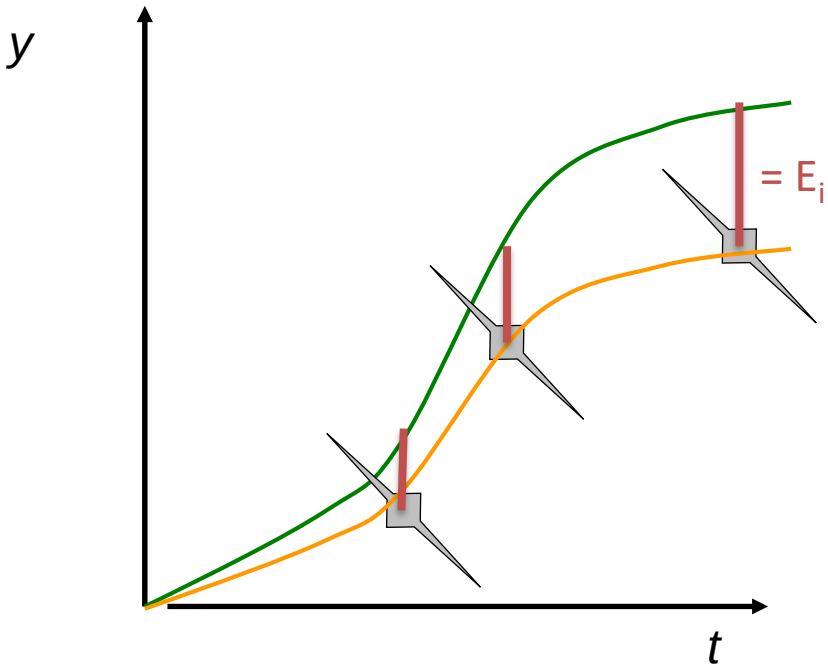
Then data selection

- Gather and qualify dataset (metadata) so that data:
  - are **representative** of the target population, i.e. the situations for which we want to make predictions (ex low N input or low SWC)
  - provide **sufficient** information and as directly **related** as possible the process of interest

  Ex. For the response of denitrification to temperature, better to have a range of representative temperature values rather than one value; better to use denitrification data than N2O data or N balance
- Don't forget that the observed data also provide imperfect information regarding the true state of the system

 observation errors should be taken into account in the calibration process (data selection, weights in the optimized criteria, …)

Calibration procedure: « classical » approach



We try to **minimize** the **sum of square** of **each value of E** :

$$\min \sum {E_i}^2$$

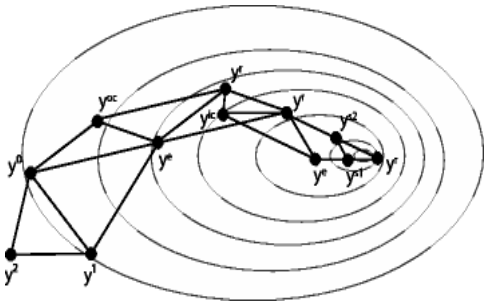
by changing the parameters values

- *Run a simulation*
- *Measure model performance (SSE)*
- *Change (efficiently) parameter value*
- *Go back to first step until SSE does not change (convergence)*

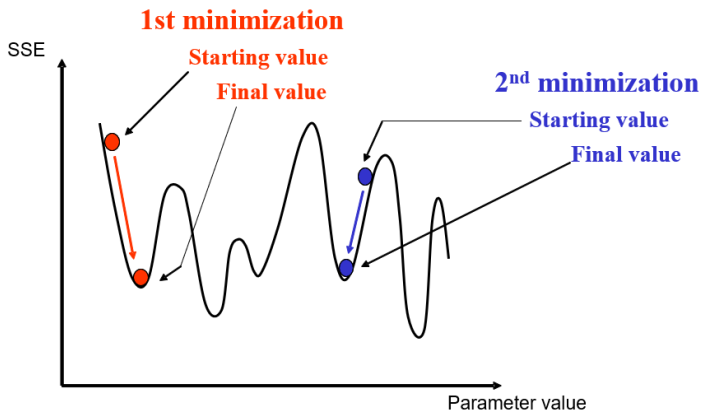
- Algorithms allow to do that **automatically**, **efficiently**, for **different parameters simultaneously**
- Different possible **criteria** (here SSE, RMSE to get same unit than y)
- Methods to take into account **prior information** (bayesian approaches)
- **Local** vs **global** optimization

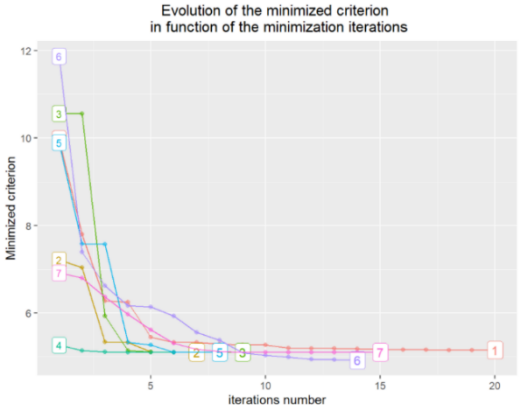
Minimization using Nelder-Mead simplex

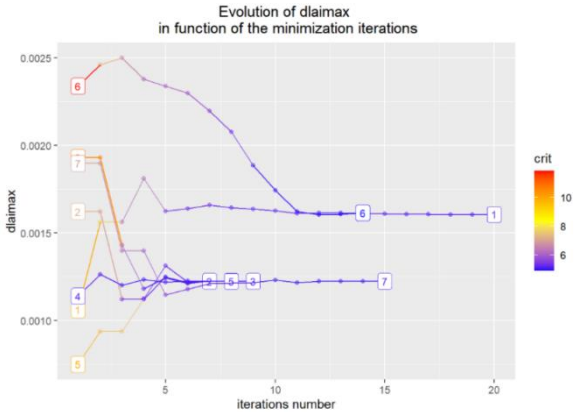
- **Iterative** algorithm
  - starting from **initial values** of the parameters to estimate
  - stopping when parameter values and/or minimized criterion **converge** (i.e. consecutive values inferior to a given threshold)



Nelder-Mead simplex minimization, from Ozaki et al, 2017







- Different starting values may lead to different results => **repeat the minimization** using different starting values
- Always better to **provide as good as possible initial values** (e.g. from expertise and/or literature)

Evaluation of the calibration

- First, check that the optimization method did its job ! (convergence, minimization of the deviations from the observations used, …)

- For a more objective/useful evaluation, evaluate the calibrated model on a dataset:
    - Representative of the target population, i.e. the situations for which we want to make predictions
    - **Not used in the calibration process**

⍰ Keep one part of the original dataset for calibration and another for evaluation (be careful with splitting)

Or

→ Use cross-validation (better but computationally expensive)

Same approach but repeated: allow to get an estimate of uncertainty on parameters (values are different at each round)

Words of caution

- Can be **manual** (trial and error) or **automatic** using an optimization algorithm and a criteria ; automatic is not always better (ex. delays)

- Only to be considered **after having carefully done baseline simulation with default parameters** + literature/measurements + analysis of the results to try to identify where the error come from

- Adjustment of parameters values should be done **within plausible range** to keep the interest of using a mechanistic model (boundary constraints, bayesian methods)

Caution (continuing)

- **Be careful to overfitting**: Estimating too many parameters with respect to the quantity of information provided (measured data) can lead to very poor predictions!

- Evaluate performances in **prediction**, not only fitting

- Be careful **to error compensation**

If you have bad N2O emissions, you may be tempted to fix it by for example changing the denitrification potential.
But, before that, you have to check that biomass growth is well simulated, mineral nitrogen dynamics is well simulated, soil water content is well simulated, ... Because if not, your changes to the denitrification potential may just try to compensate for errors in soil water content...

Last point

Your calibration depends on the state of the model at the moment; if someone changes something in the model (equation, option, parameter...), even if it does not relate to your variables of interest, it will affect the calibration !

for ex, if soil water dynamics is different, your calibration of the denitrification potential should be done again...

-> this is why the model should be reevaluated **completely** on a regular basis (done by the STICS team)

-> juxtaposition of different evaluation/calibration does not guaranty model performance!

Conclusions

- Not an easy task !
- Requires good knowledge in crop and soil functionning, measured data, model, mathematical methods and associated tools

- But we provide …
  - Software tools
  - Documentation (red book, « Guide for the adaptation of STICS to new crops »)
  - Exchange of expertise through the Stics community forum

Overall, remain critical about the limits of the calibration you (or other ones) have done!

STICS calibration: Examples

Essentially crop parameters

- Crop calibration

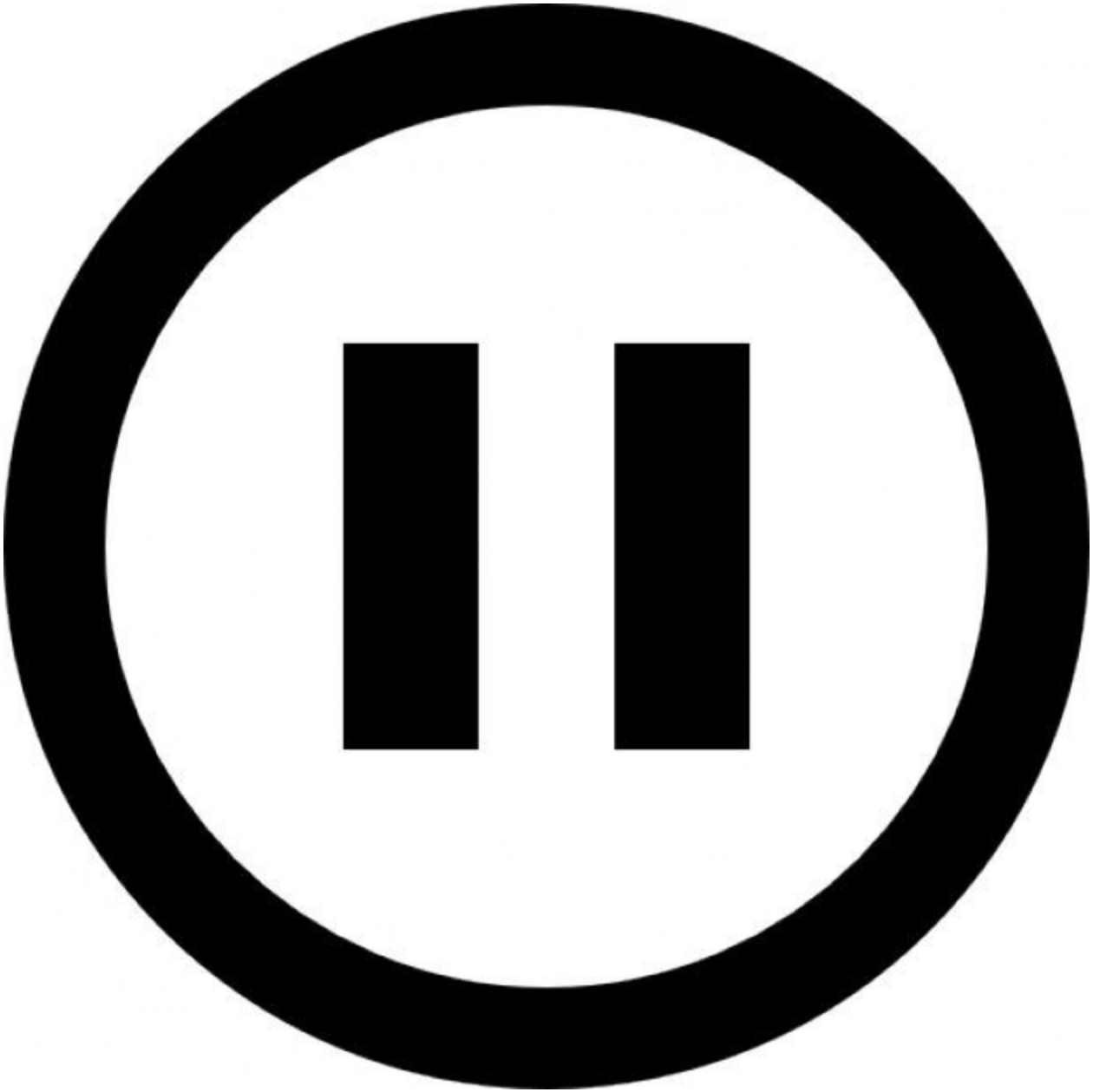
  Guillaume, S. et al (2011), Methodological comparison of calibration procedures for durum wheat parameters in the STICS model, *EJA*

  Jégo, G. et al (2013), Calibration and performance evaluation of the STICS crop model forsimulating timothy growth and nutritive value, *Field Crop Research*

- Cultivar calibration

  Jégo, G. et al (2010), Calibration and performance evaluation of soybean and spring wheat cultivars using the STICS crop model in Eastern Canada, *Field Crop Research*

  AgMIP : Wallach et al. (preprint https://www.biorxiv.org/content/10.1101/2023.10.26.564162v1.full.pdf+html), proposition of a protocol for calibration crop models, applied to STICS on synthetic experiments
  Wallach et al (2023) https://link.springer.com/article/10.1007/s13593-023-00900-0 proposition of a calibration protocol for crop phenology models and application to an ensemble of crop models



Software tools for Stics

Let’s discover them in the practical session!